The crisis arc: synchronization around global recessions

The paper's primary test. Crises are the Kose–Terrones global recessions (1975, 1982, 1991, 2009 and 2020), with the milder global downturns of 1998, 2001 and 2012 shown for contrast. Significance comes from the circular-shift permutation (53 configurations, so the smallest possible p is 1/53 = 0.019). Figures F2.1–F2.6 use the paper's convention P. The three companion figures F2.7–F2.9 were drawn with convention W, and under W the arc is not significant: see the table below and docs/open_issues.md, issue 1.

Where the Hilbert transform is applied changes the arc

The same data, filter and permutation test under three conventions (correlation, one-sided p in brackets). Recomputed 11 October 2026 by tools/check_runin_conventions.py; P reproduces kose_terrones_results.json and W reproduces amplitude_results.json exactly.

Convention	Lead-2	Lead-1	Crisis	After-1	After-2	$r-\bar{A}$
P · paper: filter and Hilbert on 1950–2022	+0.230 (0.038)	+0.216 (0.019)	+0.149 (0.189)	+0.078 (0.679)	−0.277 (0.019)	+0.608
W · filter from 1950, Hilbert on 1970–2022	+0.210 (0.057)	+0.178 (0.094)	+0.112 (0.208)	+0.088 (0.698)	−0.144 (0.170)	+0.509
N · filter and Hilbert on 1970–2022	+0.210 (0.038)	+0.178 (0.094)	+0.124 (0.189)	+0.089 (0.679)	−0.197 (0.094)	+0.531

Status key

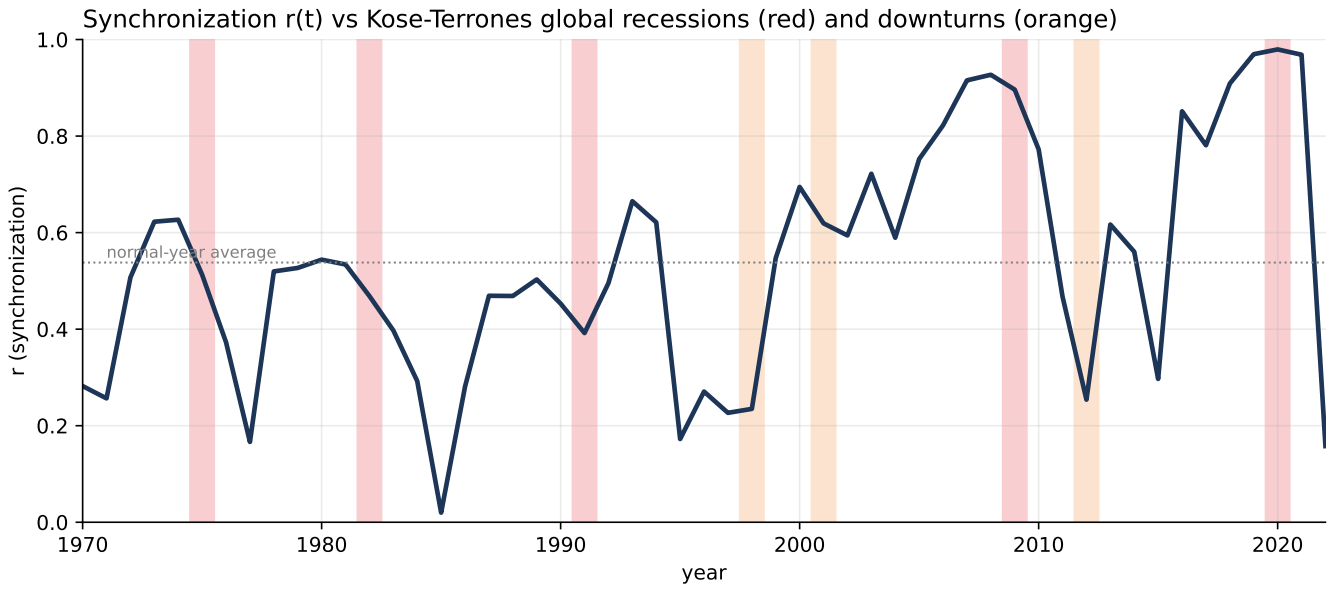
CURRENT	Paper data (Maddison 2023) and the paper's conventions; numbers match docs/methods.md.
NEEDS RELABELLING	Numbers right, but an annotation on the figure is out of date (said in the caption).
CONVENTION W	Hilbert transform applied to 1970–2022 only (convention W); the paper uses P.
SUPERSEDED	Earlier data or crisis definition; the record of how the paper got here.
DESCRIPTIVE	Atlas material from the Global Macro Database; describes, does not test.
UNVERIFIED SERIES	The series drawn matches no stored results file; redraw before use.

In this book

F2.1	Synchronization against Kose–Terrones global recessions and downturns	CURRENT
F2.2	Mean synchronization by period type	CURRENT
F2.3	Does synchronization lead global recessions?	CURRENT
F2.4	The fibrillation signature: rise, peak and shatter (superposed epoch, n = 5)	CURRENT
F2.5	The tested arc: two years before to two years after	CURRENT
F2.6	Leave-one-crisis-out: the lead is carried by 2009 and 2020	CURRENT
F2.7	Companion version: synchronization against recessions and downturns	CONVENTION W
F2.8	Companion version: rise, peak and shatter	CONVENTION W
F2.9	Companion version: the tested arc (not significant under W)	CONVENTION W

F2.1 CURRENT

Synchronization against Kose–Terrones global recessions and downturns

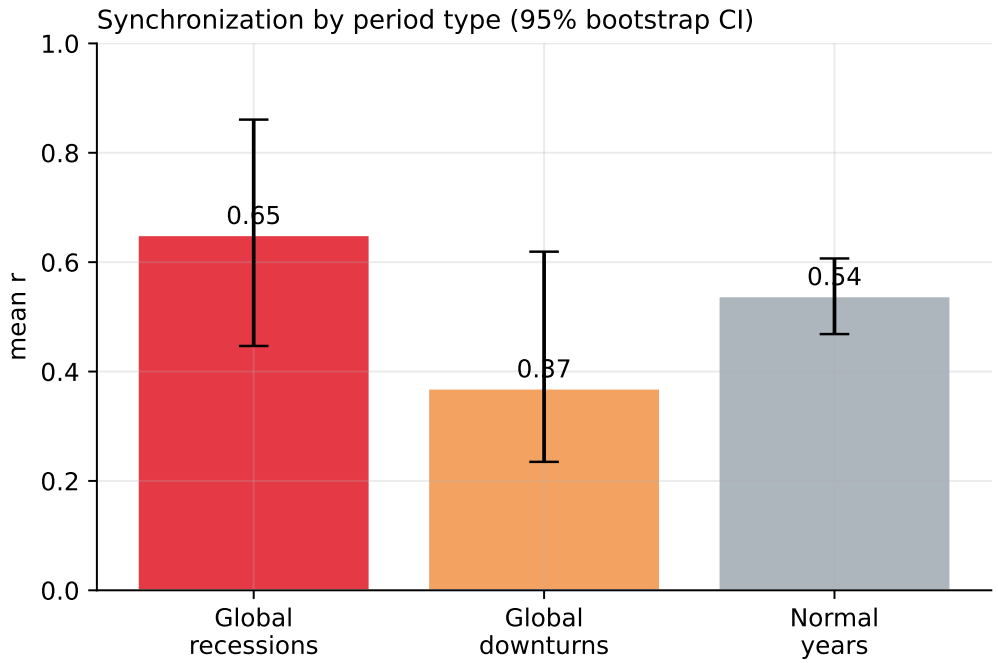


$r(t)$ for the 26 economies (convention P) with the five global recessions in red and the three global downturns in orange. The dotted line is the normal-year mean.

Numbers Normal-year mean r = 0.54.
Made by paper1/scripts/eraA_claude_ai/kose_terrones_test.py → kose_terrones_results.json (reproduces exactly) · **Data** Maddison 2023
File figures/02_crisis_arc/F2-01_sync_vs_global_recessions.pdf

F2.2 CURRENT

Mean synchronization by period type

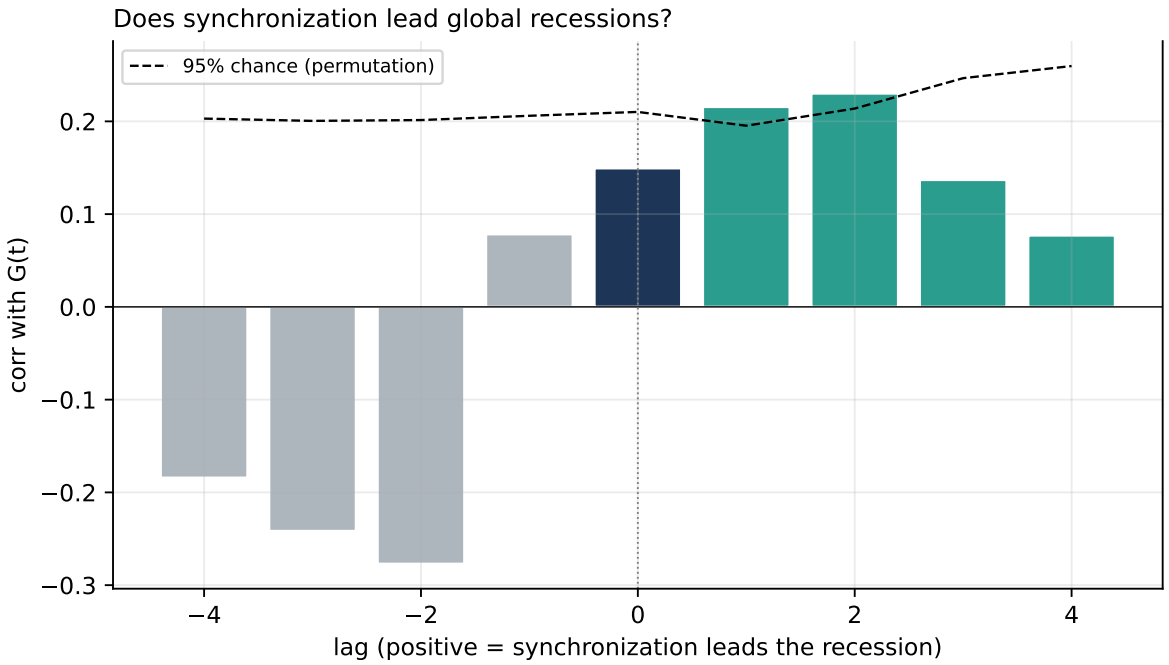


Mean r in global-recession years, global-downturn years and normal years, with 95% bootstrap intervals. Recessions are the most synchronized and milder downturns the least, below normal years. The intervals overlap, so this is description rather than a test.

Numbers Recessions 0.65 [0.45, 0.86]; downturns 0.37 [0.23, 0.62]; normal 0.54 [0.47, 0.61].
Made by kose_terrones_test.py · **Data** Maddison 2023
File figures/02_crisis_arc/F2-02_sync_by_period_type.pdf

F2.3 CURRENT

Does synchronization lead global recessions?

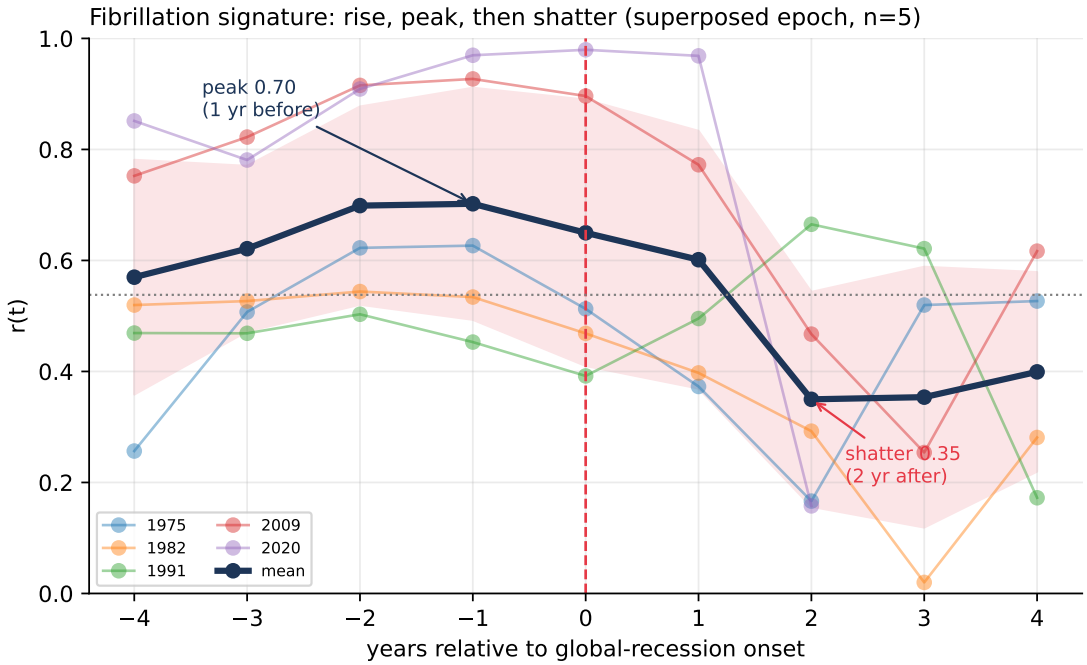


Correlation of $r(t)$ with the recession indicator shifted by -4 to $+4$ years (positive lag: synchronization comes first). The dashed line is the 95% chance level from the circular-shift permutation. The profile peaks at $+1$ and $+2$ years.

Numbers See F2.5 for the tested values.
Made by kose_terrones_test.py · **Data** Maddison 2023
File figures/02_crisis_arc/F2-03_leadlag_profile.pdf

F2.4 CURRENT

The fibrillation signature: rise, peak and shatter (superposed epoch, n = 5)

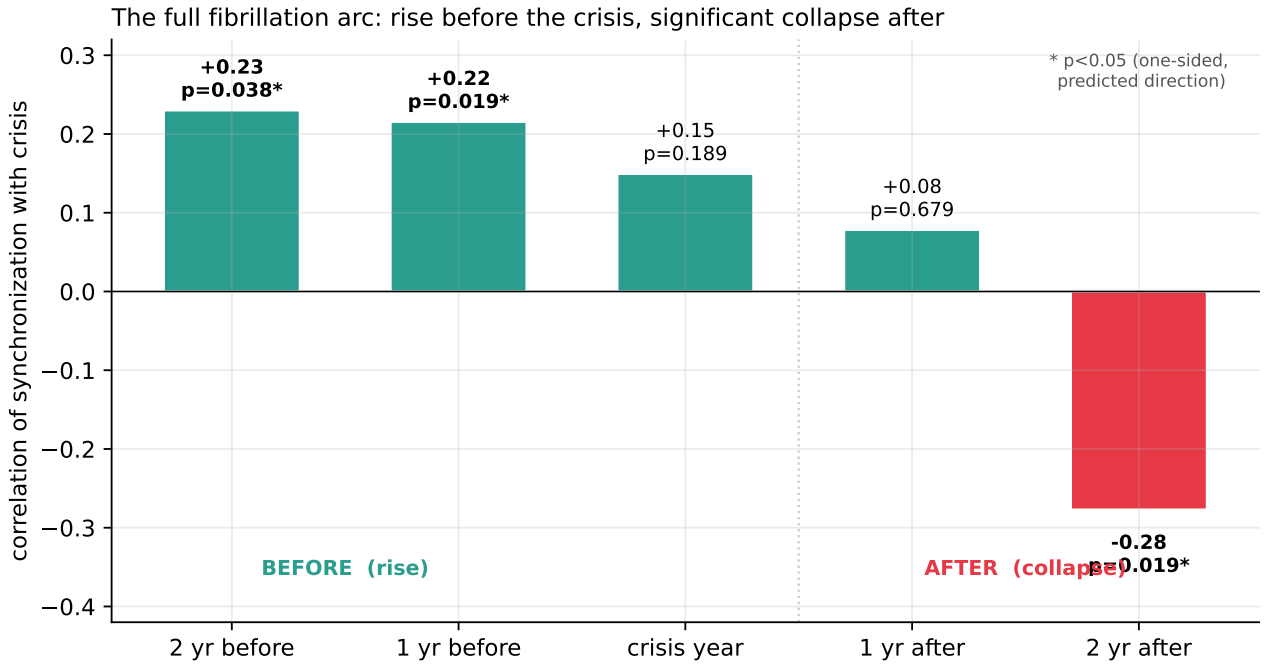


Mean r around the five recession onsets (thick line), with each recession in colour. Synchronization climbs to its peak the year before onset and collapses two years after. Descriptive alignment, no p-value.

Numbers Peak 0.70 at t−1; 0.35 at t+2; drop 0.35.
Made by kose_terrones_test.py · **Data** Maddison 2023
File figures/02_crisis_arc/F2-04_fibrillation_signature_epoch.pdf

F2.5 CURRENT

The tested arc: two years before to two years after

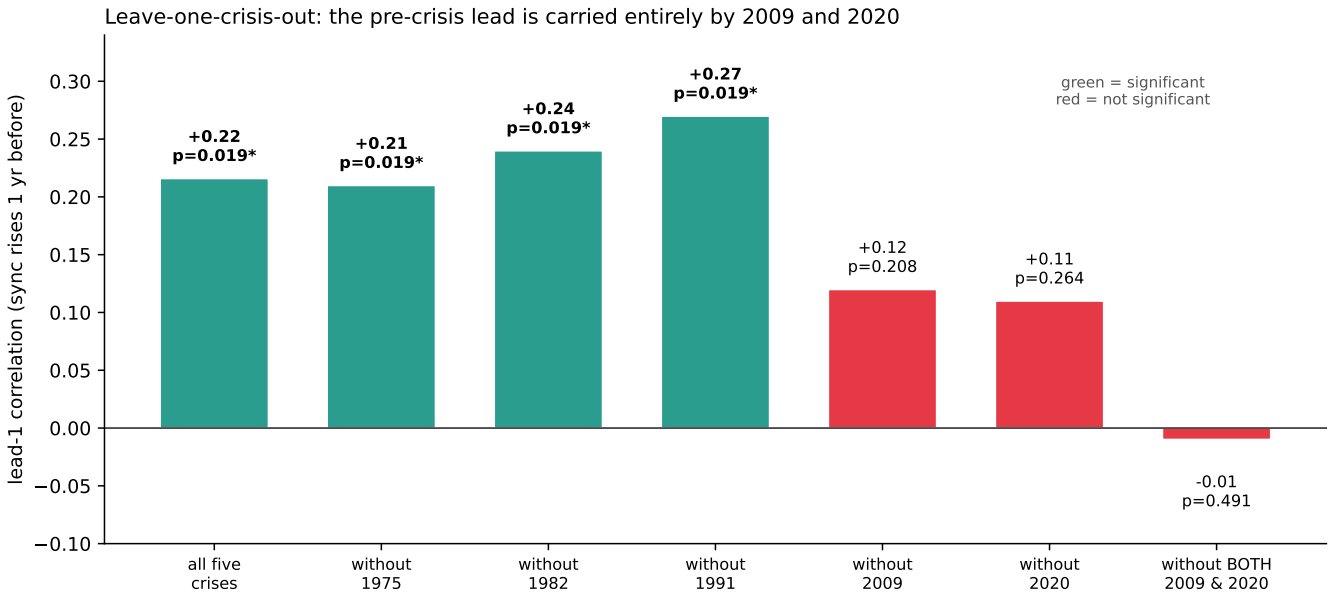


Correlation of $r(t)$ with the recession indicator at each timing, one-sided in the predicted direction. Leads and the crisis year are tested upward, lags downward. 0.019 is the permutation floor, so it means 'no shift did better', not a smaller p. Over five lags no coefficient survives Bonferroni (lead-1 and after-2: adjusted p = 0.095); Benjamini–Hochberg gives q = 0.047 for both. The arc is significant under convention P only (table above).

Numbers Lead-2 +0.23 (p = 0.038); lead-1 +0.22 (0.019); crisis year +0.15 (0.189); after-1 +0.08 (0.679); after-2 −0.28 (0.019).
Made by kose_terrones_test.py · **Data** Maddison 2023
File figures/02_crisis_arc/F2-05_fibrillation_arc_tested.pdf

F2.6 CURRENT

Leave-one-crisis-out: the lead is carried by 2009 and 2020

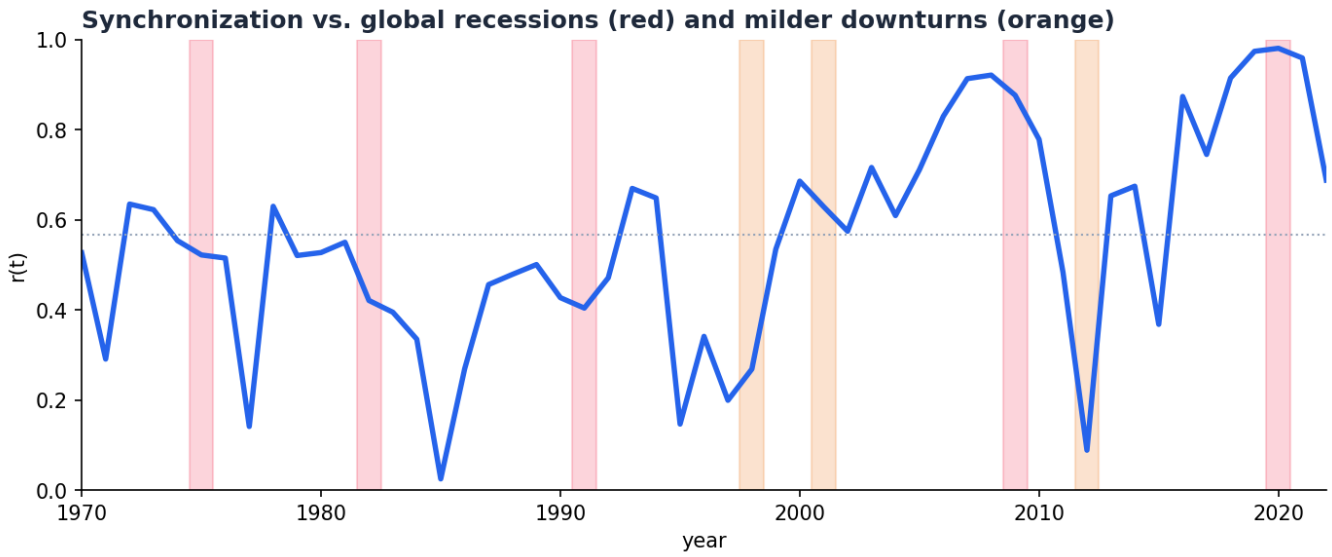


The lead-1 correlation refitted with each recession dropped in turn. It survives dropping 1975, 1982 or 1991 and disappears without 2009 or 2020. The effective event count behind the pre-crisis claim is about two, both in the modern era, which is why the paper states it as a property of today's tightly coupled economy.

Numbers All five +0.22 (0.019); without 1975 +0.21 (0.019); without 1982 +0.24 (0.019); without 1991 +0.27 (0.019); without 2009 +0.12 (0.208); without 2020 +0.11 (0.264); without both −0.01 (0.491).
Made by paper1/scripts/eraA_claude_ai/robustness.py (prints all checks; saves only the baseline) · **Data** Maddison 2023
File figures/02_crisis_arc/F2-06_leave_one_crisis_out.pdf

F2.7 CONVENTION W

Companion version: synchronization against recessions and downturns

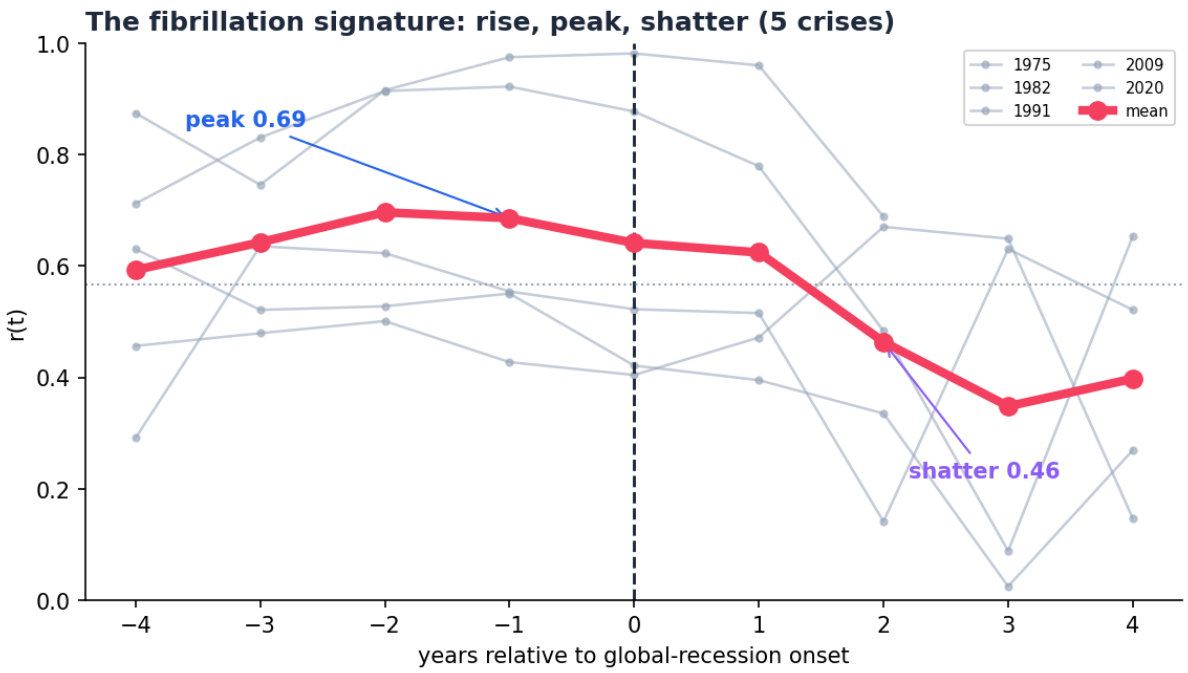


The companion book's version of F2.1. It was computed with convention W, so its $r(t)$ differs from F2.1 in many years (1970: 0.53 here against 0.28 in F2.1).

Numbers Descriptive.
Made by Companion-book figure A5 · **Data** Maddison 2023
File figures/02_crisis_arc/F2-07_companion_sync_vs_recessions.png

F2.8 CONVENTION W

Companion version: rise, peak and shatter

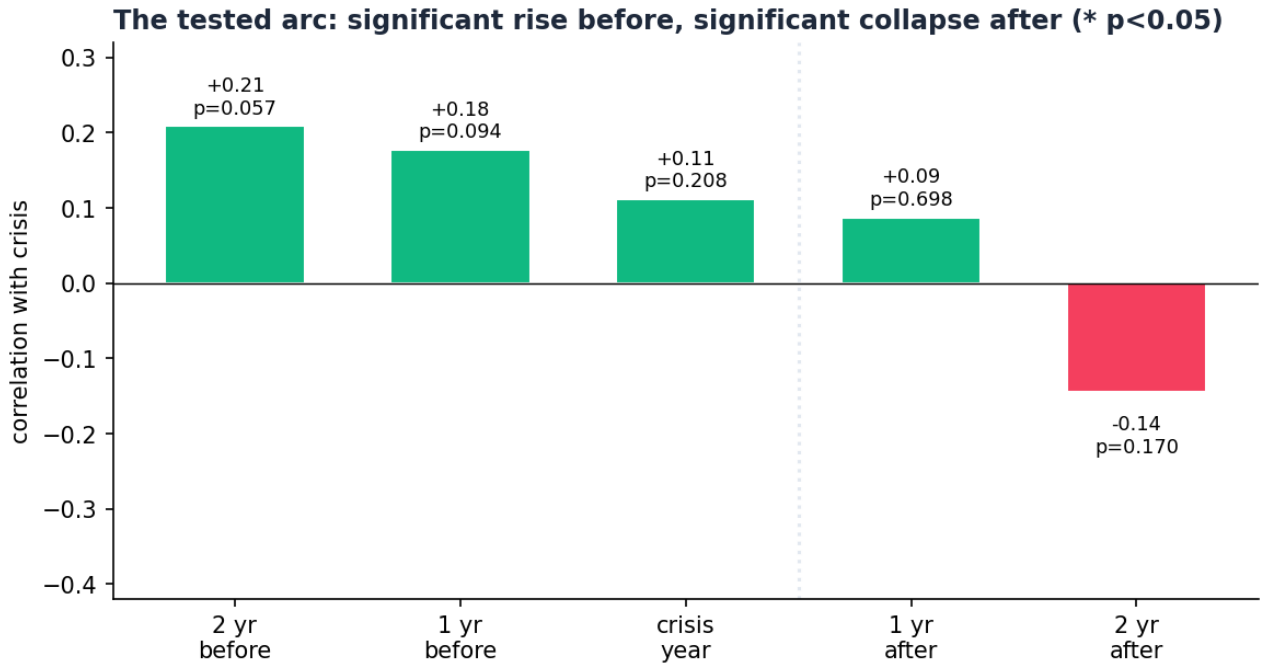


The companion book's version of F2.4, drawn with convention W. The shatter is shallower than under the paper's convention.

Numbers Peak 0.69 at t−1; 0.46 at t+2 (convention P gives 0.70 and 0.35).
Made by Companion-book figure A6 · **Data** Maddison 2023
File figures/02_crisis_arc/F2-08_companion_signature_epoch.png

F2.9 CONVENTION W

Companion version: the tested arc (not significant under W)



The companion book's version of F2.5, drawn with convention W. None of its bars reaches $p < 0.05$, although its title and the companion text say the rise and collapse are significant. Either redraw it with convention P or change the companion text; see docs/open_issues.md, issue 1.

Numbers Lead-2 +0.21 (0.057); lead-1 +0.18 (0.094); crisis year +0.11 (0.208); after-1 +0.09 (0.698); after-2 −0.14 (0.170).
Made by Companion-book figure A7 · **Data** Maddison 2023
File figures/02_crisis_arc/F2-09_companion_tested_arc.png